

Implementation Plan

DICE by Sanyog — Mobile Application

Version: 1.0.0
Date: June 2025
Status: Active Development
Current Phase: Phase 1 — MVP

1. Project Overview

DICE by Sanyog is a mobile-first certification and compliance management platform for Indian manufacturers, importers, and exporters. The implementation is divided into 4 phases across ~6 months from inception to production launch.

2. Development Phases

Phase 0	— Foundation & Setup	(Weeks 1-2)
Phase 1	— MVP Core Features	(Weeks 3-10)
Phase 2	— Advanced Features	(Weeks 11-16)
Phase 3	— Production Readiness	(Weeks 17-20)
Phase 4	— Post-Launch & Scale	(Ongoing)

3. Phase 0 — Foundation & Setup (Weeks 1-2)

Goals

Establish development environment, project structure, and CI/CD pipeline.

Deliverables

TASK	OWNER	STATUS
Expo project init (SDK 51, TypeScript)	Mobile Dev	✔ Done
React Navigation v6 structure	Mobile Dev	✔ Done
Zustand store setup (auth, theme, notifications, bookmarks)	Mobile Dev	✔ Done
ThemeContext with dark/light/system	Mobile Dev	✔ Done
Custom component library (Button, Badge, Card, Avatar)	Mobile Dev	✔ Done
App icon + splash screen (dark/light adaptive)	Mobile Dev	✔ Done
PostgreSQL schema design	Backend Dev	✔ Done
Prisma ORM setup	Backend Dev	✔ Done
Express.js / NestJS project scaffold	Backend Dev	🔄 In Progress
Git repository + branch strategy	DevOps	✔ Done
CI pipeline (GitHub Actions)	DevOps	⏸ Pending
Environment configs (.env structure)	Backend Dev	✔ Done

Branch Strategy

```
main      ← production-ready only
develop   ← integration branch
feature/* ← feature branches (merged via PR)
hotfix/*  ← critical fixes to main
```

4. Phase 1 — MVP Core Features (Weeks 3–10)

Sprint Structure: 2-week sprints

Sprint 1 (Weeks 3–4): Auth + Onboarding

Priority: P0 — Must have for any user to access the app

TASK	PRIORITY	ESTIMATE	STATUS
Login Screen UI (phone number input)	P0	1d	✅ Done
OTP Screen UI (6-digit input, resend)	P0	1d	✅ Done
Onboarding slides (3-screen carousel)	P0	1d	✅ Done
User Type Selection screen	P0	0.5d	✅ Done
Onboarding Profile screen (industries, markets)	P0	1d	✅ Done
<code>POST /auth/send-otp</code> backend	P0	0.5d	⏸ Pending
<code>POST /auth/verify-otp</code> backend	P0	0.5d	⏸ Pending
JWT + Refresh Token implementation	P0	1d	⏸ Pending
<code>PUT /user/onboarding</code> backend	P0	0.5d	⏸ Pending
useAuthStore wired to real API	P0	1d	⏸ Pending
Token storage in expo-secure-store	P0	0.5d	✅ Done
Auto-login on app restart	P0	0.5d	✅ Done

Sprint 1 Milestone: User can register, verify OTP, complete onboarding, and be auto-logged in on restart.

Sprint 2 (Weeks 5–6): Home + Navigation

Priority: P0 — Core navigation and home screen

TASK	PRIORITY	ESTIMATE	STATUS
Bottom Tab Navigator	P0	0.5d	✔ Done
Drawer Navigator (left slide)	P0	1d	✔ Done
Home Screen — greeting, stats	P0	1d	✔ Done
Home Screen — Quick Actions	P0	0.5d	✔ Done
Home Screen — Feature Carousel	P0	1d	✔ Done
Home Screen — Avatar opens drawer	P0	0.5d	✔ Done
NotificationBell component	P0	0.5d	✔ Done
Notifications Screen	P1	1d	✔ Done
GET /user/me backend	P0	0.5d	⏸ Pending
Home stats from real API	P0	0.5d	⏸ Pending
Pull-to-refresh on Home	P1	0.5d	✔ Done
AI Search bar (UI only, Phase 2 backend)	P2	0.5d	✔ Done

Sprint 2 Milestone: Home screen fully functional with real user data, navigation working across all tabs.

Sprint 3 (Weeks 7–8): Certifications + Applications

Priority: P0 — Core product value

TASK	PRIORITY	ESTIMATE	STATUS
Certifications Screen (list + filter)	P0	1d	✓ Done
Certification Detail Screen (tabs)	P0	1.5d	✓ Done
CB Comparison Screen	P1	1d	✓ Done
Certificate Center Screen	P1	1d	✓ Done
Applications Screen (list + kanban toggle)	P0	1.5d	✓ Done
Application Detail Screen (tabs)	P0	2d	✓ Done
Technical Review Screen	P0	2d	✓ Done
New Application Screen (5-step wizard)	P0	2d	✓ Done
GET /certifications backend	P0	1d	■ Pending
GET /applications backend	P0	1d	■ Pending
POST /applications backend	P0	1d	■ Pending
GET /applications/:id backend	P0	0.5d	■ Pending
Document upload (S3 presigned URLs)	P0	1d	■ Pending
GET/POST /technical-reviews backend	P0	1d	■ Pending

Sprint 3 Milestone: Users can view and create applications, view certifications, and navigate Technical Review.

Sprint 4 (Weeks 9–10): Insights + Profile + Payments

Priority: P1 — Completes core feature set

TASK	PRIORITY	ESTIMATE	STATUS
AI Insights Screen (feed + categories)	P1	1d	✓ Done
Insight Detail Screen	P1	1d	✓ Done
Bookmark functionality	P1	0.5d	✓ Done
Saved Articles Screen	P1	0.5d	✓ Done
Profile Screen	P0	1d	✓ Done
Edit Profile Screen	P0	1d	✓ Done
Security Settings Screen	P1	1d	✓ Done
Notification Settings Screen	P1	0.5d	✓ Done
Theme Settings Screen	P1	0.5d	✓ Done
Language Settings Screen	P2	0.5d	✓ Done
Payments Screen (history)	P1	1d	✓ Done
GET /insights backend	P1	1d	■ Pending
PUT /user/me backend	P0	0.5d	■ Pending
Biometric auth (expo-local-authentication)	P1	1d	■ Pending
GET/POST /payments backend	P1	1d	■ Pending
Push notification registration	P1	1d	■ Pending

Sprint 4 Milestone: Full MVP feature set complete. App is functionally complete with all screens wired to backend.

5. Phase 2 — Advanced Features (Weeks 11–16)

Sprint 5 (Weeks 11–12): AI Features

TASK	PRIORITY	ESTIMATE
AI Chat Screen (GPT-4 integration)	P1	2d
AI Assistant backend (<code>POST /ai/chat</code>)	P1	1.5d
AI Search (semantic search via embeddings)	P2	2d
AI Recommendations feed	P2	1.5d
AI-summarized circulars (GPT-4 pipeline)	P2	2d
Import/Export Alerts screen	P2	1d
Circular ingestion job (CRON)	P2	1.5d

Sprint 6 (Weeks 13–14): Documents + Shipment + Communication

TASK	PRIORITY	ESTIMATE
Documents Screen (upload, manage)	P1	1.5d
Document viewer (PDF + image)	P1	1d
Shipment Tracking Screen (map)	P2	2d
Google Maps / Mapbox integration	P2	1d
Communication Screen (in-app chat)	P1	2d
Chat backend (WebSocket or polling)	P1	2d
Support ticket creation	P2	1d
Govt Query Response from App Detail	P0	1d

Sprint 7 (Weeks 15–16): Push Notifications + Deep Links

TASK	PRIORITY	ESTIMATE
FCM integration (Android)	P0	1d
APNs integration (iOS)	P0	1d
Notification triggers (status change, expiry)	P0	2d
Deep linking setup (scheme + universal links)	P1	1.5d
In-app notification center improvements	P1	1d
Certificate expiry reminder CRON	P1	1d
Approval workflow screen	P2	1.5d

6. Phase 3 — Production Readiness (Weeks 17–20)

Sprint 8 (Weeks 17–18): Performance + Security

TASK	PRIORITY	ESTIMATE
Sentry integration (crash reporting)	P0	0.5d
API rate limiting middleware	P0	0.5d
API input validation (Zod schemas)	P0	1d
Image lazy loading + caching	P1	1d
React Query cache strategy optimization	P1	1d
Bundle size audit + optimization	P1	1d
Offline mode (read-only cached data)	P2	2d
Security audit (token storage, HTTPS, S3)	P0	1d
Remove dev bypasses (authStore temp flags)	P0	0.5d
Remove all console.log statements	P0	0.5d

Sprint 9 (Weeks 19–20): Testing + App Store

TASK	PRIORITY	ESTIMATE
E2E tests (Detox) for auth flow	P1	2d
E2E tests for application creation flow	P1	1.5d
Unit tests for stores + utilities	P1	2d
TestFlight / Internal Testing build	P0	0.5d
App Store Connect metadata (screenshots, description)	P0	1d
Google Play Console setup	P0	0.5d
App review compliance checklist	P0	0.5d
Production backend deployment (AWS ECS)	P0	1d
Database production setup (RDS)	P0	1d
CDN / CloudFront configuration	P1	0.5d

Phase 3 Milestone: App approved on TestFlight and Google Play Beta. All P0 bugs resolved.

7. Phase 4 — Post-Launch & Scale (Month 4+)

FEATURE	PRIORITY	PHASE
Hindi language support	P2	4.1
Tamil language support	P2	4.1
WhatsApp status update integration	P2	4.1
AI Certificate Finder (barcode scan)	P2	4.2
AI Product Quality Identifier	P2	4.2
Team / Multi-user accounts	P2	4.2
CB Performance Analytics dashboard	P2	4.2
Web dashboard (React)	P3	4.3
Direct BIS portal integration	P3	4.3
Subscription billing (Razorpay Subscriptions)	P1	4.1

8. Feature Priority Matrix

PRIORITY	DEFINITION	EXAMPLES
P0	Blocks app from shipping; must be in v1.0	Auth, core nav, application CRUD, cert list
P1	Important UX; ships in v1.0 or v1.1	Insights, documents, notifications, payments
P2	Nice to have; Phase 2+	AI features, shipment tracking, WhatsApp
P3	Future vision	Web dashboard, govt portal integration

9. Technical Debt Tracker

ITEM	SEVERITY	ACTION
<code>isUserTypeDone: true</code> bypass in authStore.ts	HIGH	Set to <code>false</code> before any production build
All mock data (TR_DATA, MOCK_APP, etc.)	HIGH	Replace with API calls in Sprint 3–4
Hardcoded strings in screens	LOW	Move to constants/i18n file
Missing error boundaries	MEDIUM	Add before Phase 3
No loading skeletons on lists	MEDIUM	Add shimmer placeholders in Sprint 4
Console.log statements throughout	LOW	Strip in Phase 3

10. Team Structure

ROLE	RESPONSIBILITIES
Mobile Dev Lead	React Native architecture, component library, performance
Backend Dev	Node.js API, PostgreSQL, Redis, S3, FCM/APNs
DevOps	CI/CD, AWS ECS, RDS, CloudFront, Sentry
Designer	UI/UX review, asset creation, brand consistency
QA	Test cases, E2E testing, regression testing
Product	Feature prioritization, user feedback, KPI tracking

11. Definition of Done

A feature is considered **Done** when:

- ✔ Code written and self-reviewed
- ✔ PR created and reviewed by at least one other dev
- ✔ Wired to real API (no mock data)

- 4. ☒ Both dark and light themes work correctly
- 5. ☒ Both iOS and Android tested on simulators
- 6. ☒ Empty states handled
- 7. ☒ Error states handled (network failure, API error)
- 8. ☒ Loading states handled
- 9. ☒ No TypeScript errors
- 10. ☒ No console.log in production paths
- 11. ☒ Merged to `develop` branch

12. Key Milestones

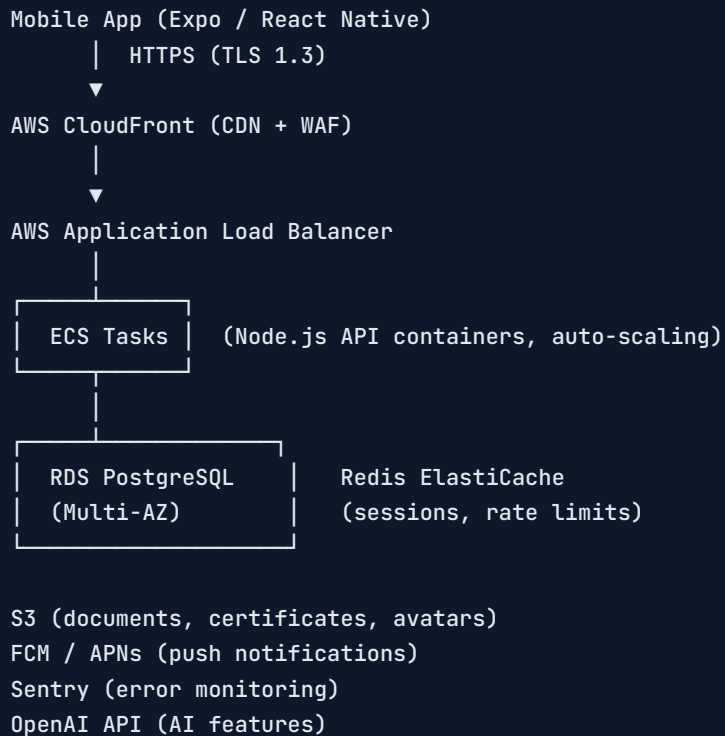
MILESTONE	TARGET DATE	CRITERIA
M0 — Dev Environment Ready	Week 2	App runs on both iOS/Android simulator
M1 — Auth Complete	Week 4	Real OTP auth working end-to-end
M2 — Core Screens Done	Week 6	All screens navigable, UI polished
M3 — Backend MVP Ready	Week 8	All P0 APIs live, app uses real data
M4 — Feature Complete	Week 10	All Phase 1 features working
M5 — Advanced Features	Week 16	AI, push, deep links, documents
M6 — TestFlight Beta	Week 18	Stable build on TestFlight
M7 — Production Launch**	Week 20	App Store + Play Store approval

13. Risk Register

RISK	PROBABILITY	IMPACT	MITIGATION
App Store rejection (privacy/content)	Medium	High	Review App Store guidelines early; include privacy policy
BIS API unavailability (no public API)	High	Medium	Build manual workflow fallback; scraping as last resort
Push notification delivery on iOS	Low	Medium	Test APNs certificate renewal; use expo-notifications abstraction
GPT-4 API cost overrun	Medium	Medium	Cache AI responses; implement usage caps per subscription tier
S3 signed URL expiry on slow uploads	Low	Low	Use multipart upload for files > 5MB
Expo SDK upgrade breaking changes	Low	Medium	Pin SDK version; upgrade in dedicated sprint
Data privacy compliance (DPDP Act)	Medium	High	Implement consent flow, data deletion endpoint, privacy policy

14. Infrastructure Architecture

Production Stack:



15. CI/CD Pipeline

```

# .github/workflows/main.yml (overview)

on:
  push:
    branches: [develop, main]
  pull_request:
    branches: [develop]

jobs:
  lint:      TypeScript typecheck + ESLint
  test:      Jest unit tests
  build-ios: eas build --platform ios --profile preview
  build-android: eas build --platform android --profile preview
  deploy-api: Docker build + push to ECR + ECS rolling deploy (main only)
  
```

16. Environment Strategy

ENVIRONMENT	PURPOSE	BACKEND URL
development	Local simulator testing with mock/real API	http://localhost:3000/v1
staging	TestFlight + Play Beta with staging backend	https://api-staging.dicebysanyog.com/v1
production	App Store release	https://api.dicebysanyog.com/v1